

Install Upgrade on Nexus Switches in vPC with NX-OS

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Introduction

This document describes the upgrade procedure of Cisco Nexus 9000 switches in vPC with Cisco NX-OS.

Prerequisites

Requirements

Cisco recommends knowledge of these topics:

- Nexus NX-OS Software.
- Virtual Port Channel (vPC).
- Use the Device File Systems, Directories, and Files on Cisco Nexus switches.
- Log into [Software Download](#).
- Cisco recommends to schedule a maintenance window since this procedure is disruptive.

Components Used

The information in this document is based on these software and hardware versions:

- Cisco Nexus 9000 with Cisco NX-OS.

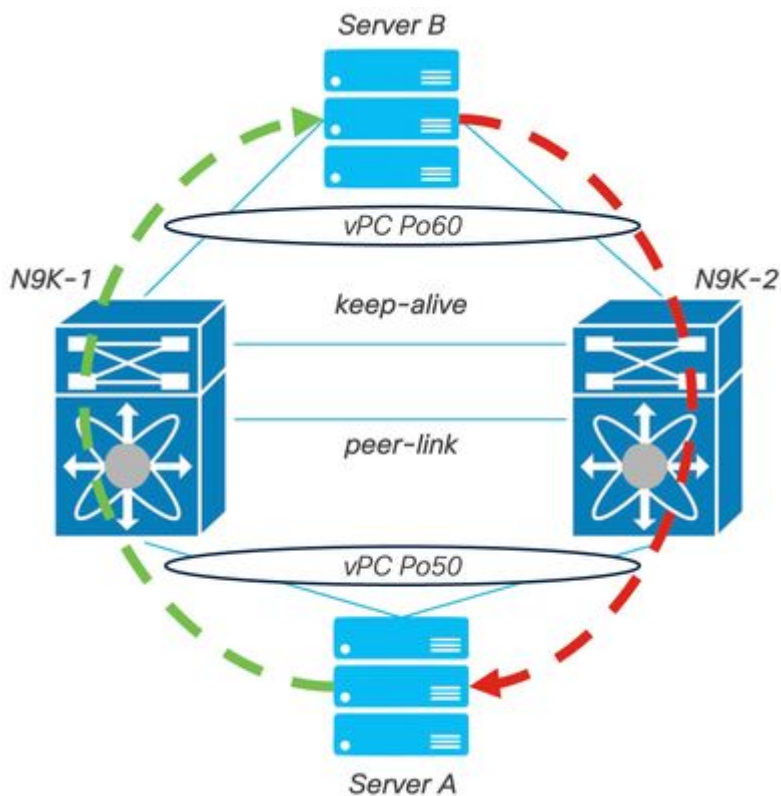


Note: The upgrade procedure for Cisco Nexus 7000 switches can be based on this document, although the commands and outputs can vary. For more information consult the official Cisco guides specific to your Cisco Nexus switch.

The information in this document was created from the devices in a specific lab environment. All of the devices used in this document started with a cleared (default) configuration. If your network is live, ensure that you understand the potential impact of any command.

Configure

Network Diagram



Network Diagram

Background Information

- Suppose you are the network administrator of a Data Center where there are two N9K-C93180YC-FX-24: N9K-1 and N9K-2 have NX-OS 9.3(11).

- Server A and B are sending production traffic.
- The goal is upgrade both Nexus switches to NX-OS 10.2(5).



Note: Start the upgrade with the primary switch, or the secondary switch does not yield any functional distinction. Nevertheless, initiating with the primary switch guarantees that both devices return to their initially configured primary and secondary roles. Although it is important to know some of the functions of the Nexus as operationally primary in vPC.

vPC Operational Primary Tasks

- Maintains ARP response functionality regardless of the FHRP configuration.
- Handles and interprets incoming BPDUs to ensure network stability.
- Facilitates the transmission of PIM multicast traffic across the network infrastructure.
- Manages and processes IGMP control-plane signaling.
- Prevents vPC member ports from transitioning to a shutdown state during a peer-link failure.

Upgrade Methodology

Step 1. Open [Cisco Nexus 9000 and 3000 ISSU Support Matrix](#)

- a. Select **Current release**.
- b. Select **Target release**.



Cisco Nexus 9000 and 3000 ISSU Support Matrix

This form provides information for In-Service Software Upgrade (ISSU) support for Cisco NX-OS software on Nexus 9000 Series and 3000 Series Switches based on your current and target release. The upgrade releases have been tested and validated by Cisco, Cisco partners, or both. Use this form as a reference for supported software.

For feedback on this tool, send email to nexus9k-docfeedback@cisco.com.

NOTE: ISSU is not supported for software downgrades. If you want to downgrade your software, follow the disruptive downgrade procedure described in the [Upgrade and Downgrade Guide](#) for your target release. For NXOS software strategy and lifecycle, see the [Cisco NX-OS Software Strategy and Lifecycle Guide](#).

Current release

Target release

Current release: 9.3(11)

Target release: 10.2(5)M

Recommended path: Direct path from Current Release. [\[Show Alternate Paths\]](#)



Note: Cisco TAC recommends that you always use disruptive (reload) upgrade option with recommended path.



Note: The recommended path can show multiple jumps. For each hop, steps 2 to 11 must be repeated until both Cisco Nexus switches have the target NX-OS version.

Step 2. Download all Cisco NX-OS files stated in recommend path depending on your Cisco Nexus switch linecard.

- Log into [Software Download](#)
- Navigate to section **Download and Upgrade** and open **Access Downloads**.
- Click **Browse all > Cisco IOS and NX-OS Software > NX-OS – NX-OS Software > Switches > Data Center Switches > Select your Nexus series > Select your linecard > NX-OS System Software > Select NX-OS version to download**.
- Move the cursor over the file name to display file details, copy the **MD5 checksum value** and download the **NX-OS file**.

Software Download

Downloads Home / IOS and NX-OS Software / NX-OS / NX-OS Software / Switches / Data Center Switches / Nexus 9000 Series Switches / Nexus 93180YC-FX-24 Switch / NX-OS System Software

Search...

Expand All

Latest Release

10.2(6)(M)

10.4(1)(F)

10.3(99x)(F)

9.3(12)

All Release

10

9

Details

Description : Cisco Nexus 9000/3000 Standalone Switch

Release : 10.2(6)

Release Date : 01-Sep-2023

FileName : nxos64-cs.10.2.6.M.bin

Min Memory : DRAM 0 Flash 0

Size : 1890.07 MB (1981878272 bytes)

MD5 Checksum : a7ab27345fb90f654a943d1765df8142

SHA512 Checksum : 4da019c09645bdf06ab78657a46c95db ...

Release Notes for 10.2(6) N3K Release Notes for 10.2(6) N9K Advisories

nxos64-cs.10.2.6.M.bin

Advisories

Related Links and Documentation

Release Notes for 10.2(6) N9K

Release Notes for 10.2(6) N3K

Release Date	Size
01-Sep-2023	1890.07 MB

Download icon

Software Download

- Transfer NX-OS files to bootflash on both Cisco Nexus switches in vPC via SCP, SFTP, TFTP or USB. If one of the first three options is selected, verify that there is a ping to the server by specifying

the expected VRF. In this example, SFTP server has IP address 192.168.9.9 reachable via Virtual Routing Forwarding (VRF) Management.

```
N9K-1(config)# ping 192.168.9.9 vrf management
```

```
N9K-1(config)# copy sftp: bootflash:
Enter source filename: nxos64-cs.10.2.5.M.bin
Enter vrf (If no input, current vrf 'default' is considered): management
Enter hostname for the sftp server: 192.168.9.9
Enter username: admin
The authenticity of host '192.168.9.9 (192.168.9.9)' can't be established.
RSA key fingerprint is SHA256:ABCDEFGHIJK.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.9.9' (RSA) to the list of known hosts.
Inbound-ReKey for 192.168.9.9
User Access Verification
Password: cisco
```

- f. Once the file transfer is complete, verify that the NX-OS files are in bootflash with command **dir**.
- g. Obtain the MD5 checksum from NX-OS file with command **show file bootflash**.
- h. Compare this value with the one copied from the [Software Download](#). Both values must match, otherwise NX-OS file is corrupted or not legit.

```
N9K-1(config)# dir | include nxos
1978203648   Mar 31 01:36:06 2023   nxos.9.3.11.bin
1943380992   Mar 17 09:54:16 2023   nxos64-cs.10.2.5.M.bin
Usage for bootflash://
20548902912 bytes used
96040308736 bytes free
116589211648 bytes total
```

```
N9K-1(config)# show file bootflash:nxos64-cs.10.2.5.M.bin md5sum
2f60a186cb9c2d55c90086302e51f655
```

Step 3. Identify the operational role in vPC for each Cisco Nexus switch.

- a. Use the command **show vpc role**.

```
N9K-1(config)# show vpc role
```

```
vPC Role status
-----
vPC role                : primary
Dual Active Detection Status : 0
vPC system-mac          : 00:23:04:ee:be:01
vPC system-priority      : 32667
vPC local system-mac     : 44:b6:be:11:17:67
```

```
vPC local role-priority      : 32667
vPC local config role-priority : 32667
vPC peer system-mac         : f8:a7:3a:4e:40:07
vPC peer role-priority      : 32667
vPC peer config role-priority : 32667
```

N9K-2(config)# show vpc role

```
vPC Role status
-----
vPC role                : secondary
Dual Active Detection Status : 0
vPC system-mac          : 00:23:04:ee:be:01
vPC system-priority     : 32667
vPC local system-mac    : f8:a7:3a:4e:40:07
vPC local role-priority : 32667
vPC local config role-priority : 32667
vPC peer system-mac     : 44:b6:be:11:17:67
vPC peer role-priority  : 32667
vPC peer config role-priority : 32667
```

Step 4. Verify incompatibility for Cisco NX-OS on both Cisco Nexus switches in vPC.

- a. Use the command **show incompatibility-all nxos bootflash**.

```
N9K-1(config)# show incompatibility-all nxos bootflash:nxos64-cs.10.2.5.M.bin
Checking incompatible configuration(s) for vdc 'N9K-1':
-----
No incompatible configurations

Checking dynamic incompatibilities:
-----
No incompatible configurations
```

Step 5. Verify the impact for Cisco NX-OS on both Cisco Nexus switches in vPC.

- a. Use the command **show install all impact nxos bootflash**. This executes a preliminary diagnosis to validate that Cisco NX-OS version is compatible and installation can be done.

```
N9K-1(config)# show install all impact nxos bootflash:nxos64-cs.10.2.5.M.bin
Installer will perform impact only check. Please wait.

Verifying image bootflash:/nxos64-cs.10.2.5.M.bin for boot variable "nxos".
```

[#####] 100% -- SUCCESS

Verifying image type.

[#####] 100% -- SUCCESS

Preparing "nxos" version info using image bootflash:/nxos64-cs.10.2.5.M.bin.

[#####] 100% -- SUCCESS

Preparing "bios" version info using image bootflash:/nxos64-cs.10.2.5.M.bin.

[#####] 100% -- SUCCESS

Performing module support checks.

[#####] 100% -- SUCCESS

Notifying services about system upgrade.

[#####] 100% -- SUCCESS

Compatibility check is done:

Module	bootable	Impact	Install-type	Reason
1	yes	disruptive	reset	default upgrade is not hitless

Images will be upgraded according to following table:

Module	Image	Running-Version(pri:alt)	New-Version	Upg-Required
1	nxos	9.3(11)		10.2(5)
1	bios	v05.47(04/28/2022):v05.43(11/22/2020)	v05.47(04/28/2022)	no

Additional info for this installation:

Service "vpc" in vdc 1: Vpc is enabled, Please make sure both Vpc peer switches have same boot mode usi

Step 6 (Optional). Export a backup of running-configuration from both Cisco Nexus switches in vPC.

```
N9K-1(config)# copy running-config sftp:running-config-backup
Enter vrf (If no input, current vrf 'default' is considered): default
Enter hostname for the sftp server: 192.168.9.9
Enter username: admin
```

```
The authenticity of host '192.168.9.9 (192.168.9.9)' can't be established.
RSA key fingerprint is SHA256:ABDCEFGHI.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.9.9' (RSA) to the list of known hosts.
Inbound-ReKey for 192.168.9.9:22
User Access Verification
Password:
Connected to 192.168.9.9.
```

Step 7. Install **NX-OS** on Nexus switch with vPC Primary role.

a. Use the command **install all nxos bootflash**.

```
N9K-1(config)# install all nxos bootflash:nxos64-cs.10.2.5.M.bin
Installer will perform compatibility check first. Please wait.
Installer is forced disruptive
```

```
Verifying image bootflash:/nxos64-cs.10.2.5.M.bin for boot variable "nxos".
[#####] 100% -- SUCCESS
```

```
Verifying image type.
[#####] 100% -- SUCCESS
```

```
Preparing "nxos" version info using image bootflash:/nxos64-cs.10.2.5.M.bin.
[#####] 100% -- SUCCESS
```

```
Preparing "bios" version info using image bootflash:/nxos64-cs.10.2.5.M.bin.
[#####] 100% -- SUCCESS
```

```
Performing module support checks.
[#####] 100% -- SUCCESS
```

```
Notifying services about system upgrade.
[#####] 100% -- SUCCESS
```

Compatibility check is done:

Module	bootable	Impact	Install-type	Reason
1	yes	disruptive	reset	default upgrade is not hitless

Images will be upgraded according to following table:

Module	Image	Running-Version(pri:alt)	New-Version	Upg-Required
1	nxos	9.3(11)		10.2(5)
1	bios	v05.47(04/28/2022):v05.43(11/22/2020)	v05.47(04/28/2022)	no

Additional info for this installation:

Service "vpc" in vdc 1: Vpc is enabled, Please make sure both Vpc peer switches have same boot mode using

Switch will be reloaded for disruptive upgrade.

Do you want to continue with the installation (y/n)? [n] y



Note: You must read success without any error. After this, Cisco Nexus switch reboots and the installation process can take several minutes. This can vary on each Cisco Nexus switch.

Step 8. Wait for the status to be active on Cisco Nexus switch.

a. Use the command **show module**.

```
N9K-1(config)# show module
Mod Ports      Module-Type      Model              Status
---
1      54      24x10/25G/32G + 6x40/100G Ethernet/FC N9K-C93180YC-FX-24  active *
```

```
Mod Sw          Hw  Slot
---
1    9.3(11)      1.0  NA
```

```
Mod  MAC-Address(es)      Serial-Num
---
1    44-b6-aa-aa-aa-aa to 44-b6-be-bb-bb-bb  ABCDEFGHIJK
```

```
Mod  Online Diag Status
---
1    Pass
```

* this terminal session



Note: Please notice it is possible that vPC keep-alive and/or peer-link cannot be in UP state. This is expected since Cisco Nexus switches in vPC have different versions of Cisco NX-OS which is unsupported.

Step 9. Install Cisco **NX-OS** on vPC peer Cisco Nexus switch.

```
N9K-2(config)# install all nxos bootflash:nxos64-cs.10.2.5.M.bin
Installer will perform compatibility check first. Please wait.
Installer is forced disruptive

Verifying image bootflash:/nxos64-cs.10.2.5.M.bin for boot variable "nxos".
[#####] 100% -- SUCCESS

Verifying image type.
[#####] 100% -- SUCCESS

Preparing "nxos" version info using image bootflash:/nxos64-cs.10.2.5.M.bin.
[#####] 100% -- SUCCESS

Preparing "bios" version info using image bootflash:/nxos64-cs.10.2.5.M.bin.
[#####] 100% -- SUCCESS

Performing module support checks.
[#####] 100% -- SUCCESS
```

Notifying services about system upgrade.
[#####] 100% -- SUCCESS

Compatibility check is done:

Module	bootable	Impact	Install-type	Reason
1	yes	disruptive	reset	default upgrade is not hitless

Images will be upgraded according to following table:

Module	Image	Running-Version(pri:alt)	New-Version	Upg-Required
1	nxos	9.3(11)		10.2(5)
1	bios	v05.47(04/28/2022):v05.42(06/14/2020)	v05.47(04/28/2022)	no

Additional info for this installation:

Service "vpc" in vdc 1: Vpc is enabled, Please make sure both Vpc peer switches have same boot mode using

Switch will be reloaded for disruptive upgrade.

Do you want to continue with the installation (y/n)? [n] y

Step 10. Wait for the status to be active on Cisco Nexus switch.

a. Use the command **show module**.

```
N9K-2(config)# show module
```

Mod	Ports	Module-Type	Model	Status
1	54	24x10/25G/32G + 6x40/100G Ethernet/FC	N9K-C93180YC-FX-24	active *

Mod	Sw	Hw	Slot
1	9.3(11)	1.0	NA

Mod	MAC-Address(es)	Serial-Num
1	f8-a7-3a-nn-nn-nn to f8-a7-3a-n1-n1-n1	98765432109

Mod	Online Diag	Status
1	Pass	

* this terminal session

Step 11. Verify the keep-alive, peer-link, and vPC port-channels are in UP state.

a. Use the command **show vpc**.

```
N9K-1(config)# show vpc
```

Legend:

(*) - local vPC is down, forwarding via vPC peer-link

```
vPC domain id          : 1
Peer status             : peer adjacency formed ok
vPC keep-alive status   : peer is alive
Configuration consistency status : success
Per-vlan consistency status : success
Type-2 consistency status : success
vPC role                : primary
Number of vPCs configured : 2
Peer Gateway            : Enabled
Dual-active excluded VLANs : -
Graceful Consistency Check : Enabled
Auto-recovery status    : Disabled
Delay-restore status    : Timer is off.(timeout = 30s)
Delay-restore SVI status : Timer is off.(timeout = 10s)
Operational Layer3 Peer-router : Enabled
Virtual-peerlink mode   : Disabled
```

vPC Peer-link status

```
-----
id   Port   Status Active vlans
--   -
1    Po1    up     1
```

vPC status

```
-----
Id   Port           Status Consistency Reason           Active vlans
--   -
50   Po50            up     success    success                1
60   Po60            up     success    success                1
```

```
N9K-2(config)# show vpc
```

Legend:

(*) - local vPC is down, forwarding via vPC peer-link

```
vPC domain id          : 1
Peer status             : peer adjacency formed ok
vPC keep-alive status   : peer is alive
Configuration consistency status : success
Per-vlan consistency status : success
Type-2 consistency status : success
vPC role                : secondary
Number of vPCs configured : 2
Peer Gateway            : Enabled
Dual-active excluded VLANs : -
Graceful Consistency Check : Enabled
```

Auto-recovery status : Disabled
Delay-restore status : Timer is off.(timeout = 30s)
Delay-restore SVI status : Timer is off.(timeout = 10s)
Operational Layer3 Peer-router : Enabled
Virtual-peerlink mode : Disabled

vPC Peer-link status

id	Port	Status	Active vlans
1	Po1	up	1

vPC status

Id	Port	Status	Consistency	Reason	Active vlans
50	Po50	up	success	success	1
60	Po60	up	success	success	1

Related Information

- [Cisco Technical Support & Downloads](#)